

# Classical Mechanics

## Final Exam

January 17th, 2024

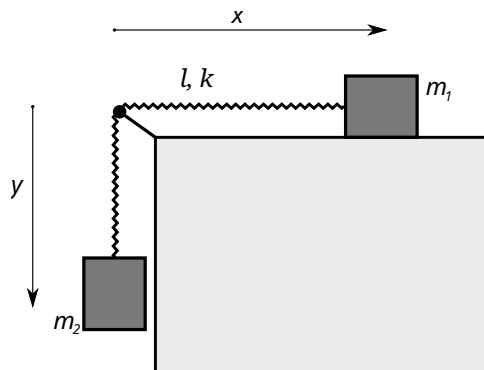
### Exercise 1 (2 points)

In a given reference frame  $S$ , event  $A$  happens at  $x = 0$  and  $t = 0$ , and event  $B$  happens at  $x = 2$  and  $t = 1$  (in units  $c = 1$ ).

- Find the frame  $S'$  (that is, the velocity of  $S'$  with respect to  $S$ ) in which the two events are simultaneous.
- Find the  $x'_B$  coordinate of event  $B$  in  $S'$ .

### Exercise 2 (3 points)

Two blocks of mass  $m$  are connected by an elastic rope as shown in the figure, so that one of the masses hangs from the edge of a table under the action of gravity, while the other slides on the surface of the table without friction. The rope has a spring constant  $k$  and natural length  $l$ .



- Write the Lagrangian of the system as a function of the  $x$  and  $y$  coordinates, as defined in the figure.
- From the Lagrangian, obtain the equations of motion for  $x$  and  $y$ .
- Write the Hamiltonian of the system. Is the Hamiltonian equal to the total energy of the system? Is the total energy conserved? Justify your answers.
- From the Hamiltonian, obtain the equations of motion for  $x$ ,  $p_x$ ,  $y$ , and  $p_y$ . Prove that they are equivalent to the equations you obtained from the Lagrangian.

### Exercise 3 (3 points)

Consider a generic Hamiltonian for a system with two degrees of freedom  $H(q_1, q_2, p_1, p_2)$ , which does not have an explicit dependency on time.

- a) Prove that, if the Hamiltonian can be written as  $H(f(q_1, p_1), g(q_2, p_2))$  with arbitrary functions  $f$  and  $g$ , then  $f$  and  $g$  are constants of motion.

Now consider, specifically,

$$H = q_1 p_1 - q_2 p_2 - a q_1^2 + b q_2^2.$$

For this Hamiltonian:

- b) Identify, by inspecting the Hamiltonian, two constants of motion.  
c) Prove that the following functions are also constants of motion

$$F_1 = \frac{p_1 - a q_1}{q_2} \quad \text{and} \quad F_2 = q_1 q_2.$$

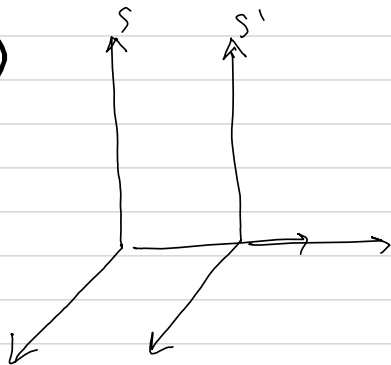
### Exercise 4 (2 points)

The Hamiltonian of a freely falling body is, in one dimension,

$$H = \frac{p^2}{2m} + mgz.$$

- a) Find a time-independent generating function  $F_4(p, P)$  such that the transformed Hamiltonian is  $\tilde{H} = P$ . (Note that there may be more than one such generating functions; you only need to find one.)  
b) Determine the explicit form of the transformation (that is, find  $Q(z, p)$  and  $P(z, p)$ ) and prove that it is canonical.

(E1)



A

B

S

$$x_A = t_A = 0$$

$$x_B = 2, t_B = 1$$

S'

$$x'_A = t'_A = 0$$

?

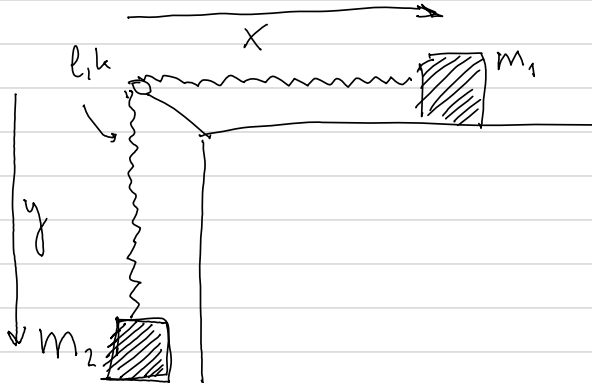
a)  $v$  such that  $t'_B = 0$  (simultaneous with A)

$$t'_B = \gamma (t_B - v x_B) \stackrel{!}{=} 0 \Leftrightarrow t_B = v x_B \Rightarrow$$

$$\Rightarrow \boxed{v} = \frac{t_B}{x_B} = \boxed{\frac{1}{2}}$$

$$b) \boxed{x'_B} = \gamma (x_B - v t_B) = \frac{2 - \frac{1}{2} \cdot 1}{\sqrt{1 - \left(\frac{1}{2}\right)^2}} = \frac{3/2}{\sqrt{3/4}} = \boxed{\sqrt{3}}$$

E2



$$a) \quad T = \frac{m_1}{2} \dot{x}^2 + \frac{m_2}{2} \dot{y}^2 \quad \left\{ \begin{array}{l} \mathcal{L} = \frac{m_1}{2} \dot{x}^2 + \frac{m_2}{2} \dot{y}^2 + m_2 g y - \frac{k}{2} (x + y - l)^2 \\ V_g = -m_2 g y \\ V_k = \frac{k}{2} (x + y - l)^2 \end{array} \right.$$

$$b) \quad \left. \begin{array}{l} \frac{\partial \mathcal{L}}{\partial x} = -k(x + y - l) \\ \frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = m_1 \ddot{x} \end{array} \right\} \Rightarrow \boxed{m_1 \ddot{x} = -k(x + y - l)}$$

$$\left. \begin{array}{l} \frac{\partial \mathcal{L}}{\partial y} = m_2 g - k(x + y - l) \\ \frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = \frac{d}{dt} (m_2 \dot{y}) = m_2 \ddot{y} \end{array} \right\} \Rightarrow \boxed{m_2 \ddot{y} = m_2 g - k(x + y - l)}$$

$$c) \quad H = p_x \dot{x} + p_y \dot{y} - \mathcal{L} \quad \left\{ \Rightarrow \right.$$

$$\left. \begin{array}{l} p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m_1 \dot{x} \quad ; \quad p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = m_2 \dot{y} \\ \Rightarrow \boxed{H = \frac{p_x^2}{m_1} + \frac{p_y^2}{m_2} - \frac{m_1}{2} \frac{p_x^2}{m_1^2} - \frac{m_2}{2} \frac{p_y^2}{m_2^2} - m_2 g y + \frac{k}{2} (x + y - l)^2} \\ \quad \quad \quad \boxed{= \frac{p_x^2}{2m_1} + \frac{p_y^2}{2m_2} - m_2 g y + \frac{k}{2} (x + y - l)^2} \end{array} \right.$$

H is the total energy because  $H = T + V$  (also because T is quadratic on the velocities), and it's conserved because  $\mathcal{L}$  does not depend explicitly on time.

$$d) \quad \dot{x} = \frac{\partial H}{\partial p_x} = \frac{p_x}{m} \quad ; \quad \dot{p}_x = -\frac{\partial H}{\partial x} = -k(x + y - l)$$

$$\Downarrow \quad p_x = \dot{x} m_1 \Rightarrow \dot{p}_x = \boxed{m_1 \ddot{x} = -k(x + y - l)} \quad \square$$

$$\dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{m_2} \quad ; \quad \dot{p}_y = -\frac{\partial H}{\partial y} = m_2 g - k(x + y - l)$$

$$\Downarrow \quad p_y = m_2 \dot{y} \Rightarrow \dot{p}_y = \boxed{m_2 \ddot{y} = m_2 g - k(x + y - l)} \quad \square$$

**E3**  $H(f(q_1, p_1), g(q_2, p_2))$

$$\begin{aligned} a) \quad \frac{df}{dt} &= [f, H] = \frac{\partial f}{\partial q_1} \frac{\partial H}{\partial p_1} - \frac{\partial H}{\partial q_1} \frac{\partial f}{\partial p_1} + \cancel{\frac{\partial f}{\partial q_2} \frac{\partial H}{\partial p_2}} - \frac{\partial H}{\partial q_2} \cancel{\frac{\partial f}{\partial p_2}} = \\ &= \frac{\partial f}{\partial q_1} \cancel{\frac{\partial H}{\partial f} \frac{\partial f}{\partial p_1}} - \frac{\partial H}{\cancel{\partial f} \frac{\partial f}{\partial q_1}} \frac{\partial f}{\partial p_1} = 0 \quad \square. \end{aligned}$$

$$\begin{aligned} \frac{dg}{dt} &= [g, H] = \cancel{\frac{\partial g}{\partial q_1} \frac{\partial H}{\partial p_1}} - \frac{\partial H}{\partial q_1} \cancel{\frac{\partial g}{\partial p_1}} + \frac{\partial g}{\partial q_2} \frac{\partial H}{\partial p_2} - \frac{\partial H}{\partial q_2} \frac{\partial g}{\partial p_2} = \\ &= \frac{\partial g}{\partial q_2} \cancel{\frac{\partial H}{\partial g} \frac{\partial g}{\partial p_2}} - \frac{\partial H}{\cancel{\partial g} \frac{\partial g}{\partial q_2}} \frac{\partial g}{\partial p_2} = 0 \end{aligned}$$

b)  $H = q_1 p_1 - q_2 p_2 - a q_1^2 + b q_2^2$

$f = q_1 p_1 - a q_1^2$  is a constant of motion

$g = -q_2 p_2 + b q_2^2$  is a constant of motion

c)  $F_1 = \frac{p_1 - a q_1}{q_2}$

$$\begin{aligned} \frac{dF_1}{dt} &= [F_1, H] = \frac{\partial F_1}{\partial q_1} \frac{\partial H}{\partial p_1} - \frac{\partial H}{\partial q_1} \frac{\partial F_1}{\partial p_1} + \frac{\partial F_1}{\partial q_2} \frac{\partial H}{\partial p_2} - \frac{\partial H}{\partial q_2} \cancel{\frac{\partial F_1}{\partial p_2}} = \\ &= \left(-\frac{a}{q_2}\right) q_1 - (p_1 - 2a q_1) \frac{1}{q_2} + \left(\frac{p_1 - a q_1}{q_2^2}\right) (-q_2) = \\ &= -\cancel{\frac{a q_1}{q_2}} - \cancel{\frac{p_1}{q_2}} + \frac{2a q_1}{q_2} + \cancel{\frac{p_1}{q_2}} - \cancel{\frac{a q_1}{q_2}} = 0 \quad \square. \end{aligned}$$

$F_2 = q_1 q_2$

$$\begin{aligned} \frac{dF_2}{dt} &= [F_2, H] = \frac{\partial F_2}{\partial q_1} \frac{\partial H}{\partial p_1} - \frac{\partial H}{\partial q_1} \cancel{\frac{\partial F_2}{\partial p_1}} + \frac{\partial F_2}{\partial q_2} \frac{\partial H}{\partial p_2} - \frac{\partial H}{\partial q_2} \cancel{\frac{\partial F_2}{\partial p_2}} = \\ &= q_2 q_1 + q_1 (-q_2) = 0 \quad \square. \end{aligned}$$

(E4)  $H = \frac{p^2}{2m} + mgz$

a)  $F_4(p, P)$  such that  $\tilde{H} = P$

Because  $H$  does not depend explicitly on time,  $\tilde{H} = H$ , so we must have that

$$P = \frac{p^2}{2m} + mgz$$

Since we are looking for  $F_4(p, P)$ , we write  $z(p, P)$

$$P = \frac{p^2}{2m} + mgz \Rightarrow \boxed{z = \frac{1}{mg} \left[ P - \frac{p^2}{2m} \right]}$$

Now, using the implicit transformation equation for  $F_4$ -type generating functions, we have:

$$\begin{aligned} z &= -\frac{\partial F_4}{\partial p} \Rightarrow F_4 = -\int z dp + f(P) = \\ &= -\int \frac{1}{mg} \left( P - \frac{p^2}{2m} \right) dp + f(P) = -\frac{Pp}{mg} + \frac{p^3}{6m} + f(P) \end{aligned}$$

Since we do not have any other restrictions, we can choose, in particular,  $f(P) = 0$ , to get

$$\boxed{F_4 = -\frac{Pp}{mg} + \frac{p^3}{6m}}$$

b) We already obtained the first transformation equation

$$\boxed{P = \frac{p^2}{2m} + mgz}$$

The second is given by the implicit transformation equation

$$\boxed{Q = \frac{\partial F_4}{\partial P} = -\frac{p}{mg}}$$

Let's now prove that the transformation is canonical:

$$[Q, P] = \cancel{\frac{\partial Q}{\partial z}} \frac{\partial P}{\partial p} - \frac{\partial P}{\partial z} \frac{\partial Q}{\partial p} = -mg \left( -\frac{1}{mg} \right) = 1 \quad \square$$